

Lesson 35: Differentiating Polynomials — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Differentiate $y = x^6$.

2. Differentiate $y = 4x^3$.

3. Differentiate $y = x^2 - 5x$.

4. Differentiate $y = 7$.

5. Differentiate $y = 3x^2 + 2x$.

6. Differentiate $y = x^5$.

Section 2 — Core practice

7. Differentiate $y = 2x^3 - 4x^2 + 7$.

8. Differentiate $y = 5x^4 - x$.

9. Differentiate $y = \sqrt{x}$ (write as a power first).

10. Differentiate $y = 1/x$.

11. Differentiate $y = x^3 + 6x - 9$.

12. Differentiate $y = 4x^5 - 3x^2 + x$.

Section 3 — Challenge & exam-style

13. $y = 2x^3 - 5x + 4$. Find dy/dx and the gradient at $x = 2$.

14. $y = x^3 - 6x^2 + 9x$. Find dy/dx and evaluate at $x = 1$.

15. Differentiate $y = (x^2 + 3)/x$ (split into terms first).

16. $y = x^4 - 2x^2$. Find the gradient at $x = -1$.